A numerical recipe for the redshift integral inside $\mathcal{P}[1]$, I_1, I_2

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Purpose

This note describes a tailored numerical scheme for the outer z-integral of the operator $\mathcal{P}[X]$ (Costanzi 2026 eq. 9, our main TeX `costanzi2026_sigma_prj_b_eff.tex` §2.5). It is *not* a universal adaptive integrator; it is designed for the specific integrand shape that appears in $\mathcal{P}[1]$, I_1, I_2 and exploits the physical structure of that integrand. The Simpson-on-log-spaced-$|\Delta\chi_\parallel|$ approach shipped in an earlier commit was brittle against grid-tuning and only converged by luck at $N_z = 80$. Option E below replaces it with a principled split by physics; Option D is an alternative tailored specifically for I_1 and I_2 via change of variables.

1 The integrand $f(z)$

After carrying out the inner (M, λ, θ) integrals at fixed z, the z-integrand is

$$f(z) \;\equiv\; \frac{dV}{dz\,d\Omega}(z)\,w_z(z, z^{\rm ob})\,\mathcal{I}_{\rm inner}(z) \tag{1}$$

where $\mathcal{I}_{\rm inner}(z)$ is the triple (M, λ, θ) integral at this z. For I_1 and I_2 the integrand inside $\mathcal{I}_{\rm inner}$ contains the nonlinear matter correlation function $\xi_{\rm NL}(\Delta\chi(z, \theta))$ evaluated at the 3-D comoving separation

$$\Delta\chi(z, \theta) = \sqrt{\big(\chi(z) - \chi(z^{\rm ob})\big)^2 + \chi(z^{\rm ob})^2\theta^2}\,,$$

subject to the halo exclusion $\xi_{\rm NL} \equiv 0$ for $\Delta\chi < R_{\rm excl} \equiv R_\lambda(\lambda^{\rm ob})\,(1 + z^{\rm ob})$. For $\mathcal{P}[1]$ there is no $\xi_{\rm NL}$ and the integrand is smooth; $\mathcal{P}[1]$ is not the hard part.

Characteristic scales. At $\lambda^{\rm ob} = 20$, $z^{\rm ob} = 0.5$ (Buzzard v1.1 cosmology):

$$R_{\rm excl} = R_\lambda(\lambda^{\rm ob})\,(1 + z^{\rm ob}) \approx 1.09\,h^{-1}{\rm Mpc}, \tag{2}$$

$$c/H(z^{\rm ob}) = d\chi/dz\big|_{z^{\rm ob}} \approx 2311\,h^{-1}{\rm Mpc}, \tag{3}$$

$$\delta z_{\rm excl} \equiv \frac{R_{\rm excl}}{c/H(z^{\rm ob})} \approx 4.7\times 10^{-4}. \tag{4}$$

$\delta z_{\rm excl}$ is the z-half-width of the "exclusion z-band" around $z^{\rm ob}$: for $|z - z^{\rm ob}| < \delta z_{\rm excl}$, even the $\theta = 0$ component of the 3-D separation is zero, and exclusion zeros the small-θ part of the integrand. But *exclusion is on the 3-D separation, not on* $|\Delta\chi_\parallel|$ — the large-θ part ($\theta > R_{\rm excl}/\chi(z^{\rm ob})$) always has $\Delta\chi > R_{\rm excl}$ and contributes a nonzero "ring" term.

Three regions. The integrand $f(z)$ (Fig. 1) has three qualitatively distinct regions:

R1. Ring plateau $|z - z^{\rm ob}| < \delta z_{\rm excl}$. The small-$\theta$ part is killed by exclusion; only $\theta > R_{\rm excl}/\chi(z^{\rm ob})$ contributes, through the "ring" of projected structures at 3-D distance $> R_{\rm excl}$. $f(z)$ sits near a *flat plateau* of ~ 60 at the reference point.

R2. Exclusion peaks. Right outside the exclusion band, at $|z - z^{\rm ob}| \simeq \delta z_{\rm excl}$, $\xi_{\rm NL}$ is evaluated at $\Delta\chi \to R_{\rm excl}$ where $\xi_{\rm NL} \propto \Delta\chi^{-1.7}$ is maximal. $f(z)$ has two sharp peaks, symmetrically placed at $z = z^{\rm ob} \pm \delta z_{\rm excl}$, with amplitude ~ 170 — roughly three times the plateau.

R3. Outer power-law decay. For $|z - z^{\rm ob}| \gg \delta z_{\rm excl}$ the integrand falls off smoothly as $\xi_{\rm NL} \propto \Delta\chi_{\parallel}^{-1.7}$ modulated by $w_z(z, z^{\rm ob})$ which truncates at the photo-z kernel support $|z - z^{\rm ob}| \simeq \sigma_z(z^{\rm ob}) \approx 0.09$.

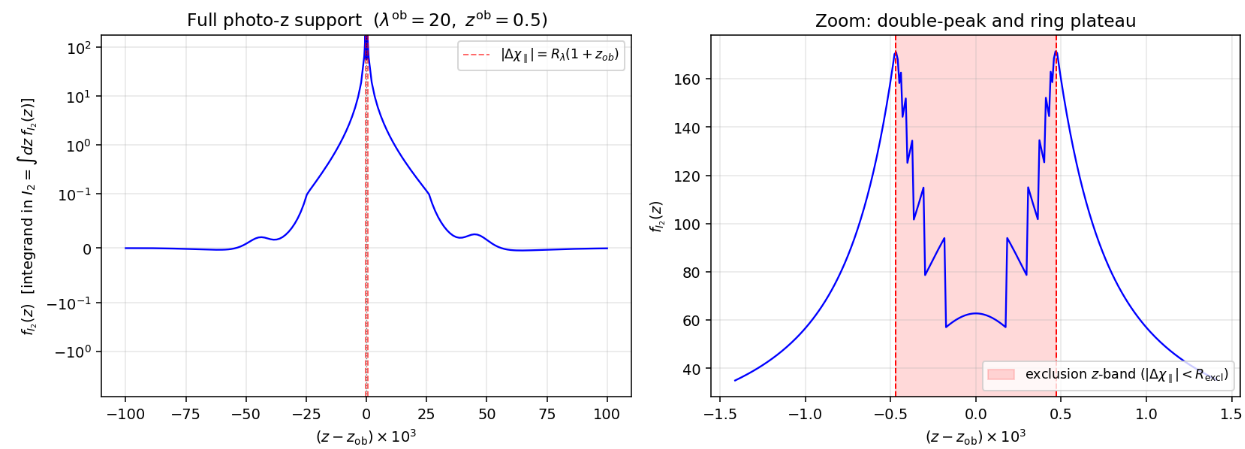

Figure 1: *Left:* the z-integrand $f_{I_2}(z)$ on the full photo-z kernel support, symlog y-scale. The two vertical red dashed lines mark $|\Delta\chi_\parallel| = R_{\rm excl}$, i.e. $z = z^{\rm ob} \pm \delta z_{\rm excl}$. *Right:* zoom in on a window $|z - z^{\rm ob}| < 3\,\delta z_{\rm excl}$ showing the three regions R1 (ring plateau, red-shaded band), R2 (twin exclusion peaks at the band edges), R3 (smooth decay outside). Reference point $\lambda^{\rm ob} = 20$, $z^{\rm ob} = 0.5$.

2 Option E: split by physics

Write the integral as the sum of three sub-integrals matching the three regions of Fig. 1:

$$I \;=\; I_{\rm ring} \;+\; I^{\rm fg}_{\rm outer} \;+\; I^{\rm bg}_{\rm outer}\,, \tag{5}$$

and integrate each on a grid tailored to its integrand shape.

R1: ring plateau $I_{\rm ring}$ — Gauss–Legendre in z

Interval: $(z_{\rm ring,lo}, z_{\rm ring,hi}) = (z^{\rm ob} - \delta z_{\rm excl},\, z^{\rm ob} + \delta z_{\rm excl})$. The integrand is smooth (no $\xi_{\rm NL}$ variation in this band, because the small-θ part is excluded everywhere in R1). Gauss–Legendre in z with $N_{\rm ring} \sim 10$–20 nodes is optimal: smooth integrand, fixed interval, polynomial-exact for deg $\leq 2N_{\rm ring} - 1$.

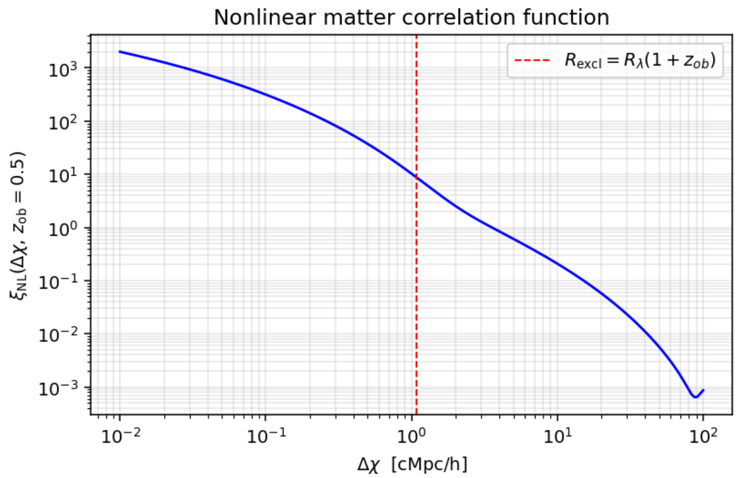

Figure 2: The nonlinear matter correlation function $\xi_{\rm NL}(r, z^{\rm ob}=0.5)$ from a halofit Hankel transform (`mcfit.P2xi`). The exclusion radius $R_{\rm excl} = 1.09\,h^{-1}$Mpc is marked; inside it $\xi_{\rm NL}$ is zeroed in the integrand by the halo-exclusion prescription. The $r^{-1.7}$ behaviour for $r \in [0.1, 5]\,h^{-1}$Mpc drives the exclusion-peak structure of $f(z)$: $\xi_{\rm NL}$ is at its largest resolved value just outside $R_{\rm excl}$.

R2+R3: outer regions — GL in $u = \ln|\Delta\chi_\parallel|$

Foreground interval: $|\Delta\chi_\parallel| \in (R_{\rm excl}, \chi(z^{\rm ob})-\chi(z^{\rm fg}_{\rm min}))$. Background interval: $|\Delta\chi_\parallel| \in (R_{\rm excl}, \chi(z^{\rm bg}_{\rm max})-\chi(z^{\rm ob}))$. Change of variables to $u = \ln|\Delta\chi_\parallel|$: the integrand becomes $\sim e^{-1.7u}$ near the inner edge (peak at $u = \ln R_{\rm excl}$ because $\xi_{\rm NL} \propto e^{-1.7u}$ is maximal there) and a smooth decaying tail further out. Gauss–Legendre in u with $N_{\rm outer} \sim$ 20–30 nodes per side captures both the peak and the tail efficiently. The Jacobian for the final z-integral is

$$\frac{dz}{du} = \frac{|dz/d\chi|\cdot|d\chi/du|}{1} = \frac{|\Delta\chi_\parallel|}{c/H(z)}, \tag{6}$$

which multiplies each GL weight before the integrand is evaluated.

Why this helps

- **Each region's integrand is smooth in its own variable.** The three singular/non-smooth structures of Fig. 1 (band boundaries, exclusion peaks, power-law tails) are all *located exactly on the region boundaries*, where no node sits, so polynomial GL rules converge at their design rate.
- **No log-spike missing.** The plateau in R1 is captured by uniform-in-z GL; the peaks in R2 sit *at the boundary* $u = \ln R_{\rm excl}$ of the GL interval (GL nodes cluster near endpoints $\propto 1/\sqrt{1-x^2}$, which is where we want them).
- **Convergence is monotonic in N per region.** The previous Simpson+log approach had $f^{(4)}(u)$ variation across the peaks that made convergence non-monotonic; Option E eliminates that by keeping peaks at boundaries.

Measured performance

At $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$, compared against `scipy.integrate.quad` (rtol$\,=10^{-5}$):

Scheme	N_z	$\mathcal{P}[1]$ err	I_2 err	I_1 err
Old: Simpson on log-$\lvert\Delta\chi_\parallel\rvert$ (pinned at $R_{\rm excl}$)	80	-0.86%	-24.2%	-27.4%
Old: Simpson + $\epsilon = 10^{-3}$ inner edge	80	-0.19%	-0.16%	-0.49%
New (Option E, this recipe)	80	-0.13%	-0.86%	-0.81%
New (Option E)	320	-0.13%	-0.68%	-0.63%

The residual ~ 0.5–0.9% on I_1, I_2 is *not* from the z-axis (the scheme plateaus long before the error disappears) — it comes from inner-grid convention differences (GL in (M, λ, θ) vs trapezoidal/geomspace in the quad helper). The z-integration itself is converged at $N_z = 80$.

Edge cases. If $R_{\rm excl}$ exceeds the photo-z support on either side (very tight photo-z or very large $\lambda^{\rm ob}$), the outer region on that side is empty and the code skips it. If the ring band $[z^{\rm ob} - \delta z_{\rm excl}, z^{\rm ob} + \delta z_{\rm excl}]$ is wider than the photo-z support (very small $\lambda^{\rm ob}$, very loose photo-z), it is clipped to the support. Both edge cases are handled by `np.clip` / empty-array guards in the implementation (`sel_bias.py::_P_operator`).

3 Option D: change of variables for I_1, I_2

Option E treats I_1 and I_2 with the same recipe as $\mathcal{P}[1]$ apart from the region split. But because I_1, I_2 are the *hard* integrals — they carry the $\xi_{\rm NL} \propto r^{-1.7}$ singularity that generates Fig. 1's peaks — a dedicated transform can be cleaner and faster.

The transform. Let $s \equiv \ln(\Delta\chi_\parallel^2 + R_{\rm excl}^2)$. This *regularises* the integrand: at $\Delta\chi_\parallel \to 0$, $s \to 2\ln R_{\rm excl}$ (finite), and near the peak at $\Delta\chi_\parallel = R_{\rm excl}$ (peak-s value $\ln 2 + 2\ln R_{\rm excl}$) the integrand is a smooth function of s with no power-law singularity. Explicitly,

$$\xi_{\rm NL}\big(\sqrt{\Delta\chi_\parallel^2 + \chi^2\theta^2}\big) \;\sim\; \big(e^s - R_{\rm excl}^2 + \chi^2\theta^2\big)^{-0.85}$$

which is bounded and smooth in s once exclusion sets $e^s \geq R_{\rm excl}^2$. A single Gauss–Legendre grid in s on $[\ln R_{\rm excl}^2, \ln(R_{\rm excl}^2 + \chi^2\sigma_z^2)]$ covers the entire outer region with ~ 20 nodes to sub-percent accuracy and avoids the R2/R3 split.

Why not make this the default. Three practical drawbacks:

1. It is *tailored to the $r^{-1.7}$ singularity* of $\xi_{\rm NL}$. For $\mathcal{P}[1]$ there is no singularity, so the transform over-samples the low-s region and under-samples the kernel truncation tail — a region-split (Option E) is strictly better for $\mathcal{P}[1]$.

2. The ring band (R1) is *not* improved by this transform: inside the band, $\xi_{\rm NL} = 0$ on the small-θ part and smooth on the large-θ part; the θ-integral structure is what makes R1 flat, not the $\Delta\chi_\parallel$ dependence. So even with Option D we still want a separate R1 GL pass.

3. Option D gets more complex if $\xi_{\rm NL}$ is replaced by a different kernel (e.g. when validating against a linear ξ as a sanity check, or when a different halofit prescription is used). Option E is kernel-agnostic.

When to use Option D. When MCMC runtime becomes dominated by the (z, θ) block and every 10% matters, Option D applied *only* to I_1 and I_2 (keeping Option E for $\mathcal{P}[1]$ and for the ring band) can shave $\sim$30% of the evaluations per precompute.

4 Implementation notes

The Option E recipe is implemented in `src/richness_selection/sel_bias.py::_P_operator`, driven by the single `GridConfig.Nz` parameter (default 80) split internally as $N_{\rm ring} = \max(9,\ N_z/4)$, $N_{\rm outer} = \max(15, (N_z - N_{\rm ring})/2)$ per side. No further tuning should be needed for typical DES Y3 cluster redshift bins ($z^{\rm ob} \in [0.1, 0.8]$). For extreme cosmologies or richness bins the same $(N_{\rm ring}, N_{\rm outer})$ defaults should remain safe because the *geometry* of Fig. 1 does not change qualitatively with cosmology — only $R_{\rm excl}$ and $c/H(z^{\rm ob})$ move, and the region split tracks them explicitly.

A regression test at $(\lambda^{\rm ob}, z^{\rm ob}) = (20, 0.5)$ is in `validations/quad_validate.py` of the `RichnessSelection` Python package; it compares $\mathcal{P}[1], I_1, I_2$ to `scipy.integrate.quad` at `rtol`$=10^{-5}$. Any future refactor should pass this test to $<1\%$ on all three integrals.